[443] Note on the Solution of the Quartic Equation $aU + 6\beta H = 0$ (concluded).

so that identically $X^2 + Y^2 + Z^2 = 0$, the expression $\alpha X + \beta Y + \gamma Z$ will be a square if only $\alpha^2 + \beta^2 + \gamma^2 = 0$. (To see this observe that in virtue of the equation $X^2 + Y^2 + Z^2 = 0$, we have $X + iY$, $X - iY$ each of them a square, and thence

$$\alpha X + \beta Y + \gamma Z = \tfrac{1}{2}(\alpha + i\beta)(X - iY) + \tfrac{1}{2}(\alpha - i\beta)(X + iY) - \gamma i \sqrt{X^2 + Y^2}$$

is a square if the condition in question be satisfied.)

Hence in particular writing

$$\sqrt{\omega_2 - \omega_3}\,\sqrt{aJ + 6\beta\omega_1 J},\ \ldots,\ \sqrt{\omega_1 - \omega_2}\,\sqrt{aJ + 6\beta\omega_3 J},$$

for α, β, γ, we have

$$(\omega_2 - \omega_3)\sqrt{aJ + 6\beta\omega_1 J}\,\sqrt{H} + (\omega_3 - \omega_1)\sqrt{aJ + 6\beta\omega_2 J}\,\sqrt{H} + (\omega_1 - \omega_2)\sqrt{aJ + 6\beta\omega_3 J}\,\sqrt{H}$$

a perfect square, and since the product of the four different values is a multiple of $(aU + 6\beta H)^2$ (this is most readily seen by observing that for $aU + 6\beta H = 0$, the irrational expression omitting a factor is

$$(\omega_2 - \omega_3)(aJ + 6\beta\omega_1 J) + (\omega_3 - \omega_1)(aJ + 6\beta\omega_2 J) + (\omega_1 - \omega_2)(aJ + 6\beta\omega_3 J),$$

which vanishes identically) it follows that the expression in question is the square of a linear factor of $aU + 6\beta H$.

It thus appears that the radicals (other than those arising from the solution of $U = 0$) contained in the solution of the equation $aU + 6\beta H = 0$ are the three roots

$$\sqrt{aJ + 6\beta\omega_1 J},\quad \sqrt{aJ + 6\beta\omega_2 J},\quad \sqrt{aJ + 6\beta\omega_3 J}.$$

Cambridge, September 2, 1868.

[444] On the Centro-surface of an Ellipsoid.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 16–18.]

The President [Prof. Cayley] gave an account of his investigations on the centro-surface of an ellipsoid (locus of the centres of curvature of the ellipsoid). The surface has been studied by Dr Salmon, and also by Prof. Clebsch, but in particular the theory of the nodal curve on the surface admits of further development. The position of a point on the ellipsoid is determined by means of the parameters, or elliptic coordinates, h, k; viz., if as usual a, b, c are the semi-axes, and if X, Y, Z are the coordinates of the point in question, then

$$\frac{X^2}{a^2 + h} + \frac{Y^2}{b^2 + h} + \frac{Z^2}{c^2 + h} = 1,$$

$$\frac{X^2}{a^2 + k} + \frac{Y^2}{b^2 + k} + \frac{Z^2}{c^2 + k} = 1;$$

and hence

$$-\beta\gamma X^2 = a^2(a^2 + h)(a^2 + k),$$

$$-\gamma\alpha Y^2 = b^2(b^2+h)(b^2+k),$$
$$-\alpha\beta Z^2 = c^2(c^2+h)(c^2+k),$$

if for shortness

$$\alpha = b^2 - c^2, \quad \beta = c^2 - a^2, \quad \gamma = a^2 - b^2 \qquad (\alpha+\beta+\gamma = 0).$$

This being so, the coordinates of the point of intersection of the normal at (X, Y, Z) by the normal at the consecutive point of the curve of curvature

$$\frac{X^2}{a^2+k} + \frac{Y^2}{b^2+k} + \frac{Z^2}{c^2+k} = 1$$

are given by the formulae

$$-\beta\gamma a^2x^2 = (a^2+h)^2(a^2+k),$$
$$-\gamma\alpha b^2y^2 = (b^2+h)^2(b^2+k),$$
$$-\alpha\beta c^2z^2 = (c^2+h)^2(c^2+k);$$

viz., these equations, considering therein (h, k) as arbitrary parameters, determine the coordinates (x, y, z) of a point on the centre-surface. The principal sections (as is known) consist each of them of an ellipse counting three times, and of an evolute of an ellipse; the evolute and ellipse have four contacts (two-fold intersections) and four simple intersections, but the contacts and intersections respectively are in the different sections real and imaginary; and if (as we may without loss of generality assume) $a^2 + c^2 > 2b^2$, then the form of the principal sections is as shown in the figure (which represents only an octant of the surface); viz., there is a real contact at P in the plane of xz, and a real intersection at Q in the plane of xy. The surface has thus an exterior and an interior sheet, but (instead of meeting in a conical point, as in the wave surface) these intersect in a nodal curve QP. The curve has a cusp at Q, and a node at P; viz., the curve extends beyond P, but from that point is acnodal, or without any real sheet of the surface passing through it. For the nodal curve there must be two values (h, k), (h_1, k_1), giving the same values of (x, y, z); viz., there must exist the relations

$$(a^2+h)^2(a^2+k) = (a^2+h_1)^2(a^2+k_1),$$
$$(b^2+h)^2(b^2+k) = (b^2+h_1)^2(b^2+k_1),$$
$$(c^2+h)^2(c^2+k) = (c^2+h_1)^2(c^2+k_1);$$

from which equations eliminating k and k_1 we should have between h, k a relation which, combined with the expressions of x, y, z in terms of (h, k), determines the nodal curve. But the better course is to eliminate k, k_1, thus obtaining a relation between h and h_1, in virtue whereof h_1 may be regarded as a known function of h; k and k_1 can then be readily expressed in terms of h, h_1; that is, we have k as a function of h, h_1, or in effect as a function of h. The relation between h, h_1 (after a reduction of some complexity) assumes ultimately a form which is very simple and remarkable; viz., writing

$$P = a^2 + b^2 + c^2, \quad Q = b^2c^2 + c^2a^2 + a^2b^2, \quad R = a^2b^2c^2,$$

the relation is

$$\begin{aligned}&(QR + 3Qh^2 + Ph^4)\\ &+ h_1(3Qh + 4Ph^2 + 3h^3)\\ &+ h_1^2(P + 3h^2) = 0;\end{aligned}$$

this is a $(2, 2)$ correspondence between the two parameters h, h_1; the united values $h_1 = h$, are given by the equation $6(R + Qh + Ph^2 + h^3) = 0$, that is

$$(a^2 + h)(b^2 + h)(c^2 + h) = 0;$$

viz., the two points on the ellipsoid which have their common centre of curvature on the nodal curve are only situate on the same curve of curvature when this curve is a principal section of the ellipsoid.

[Since the date of the foregoing communication, Prof. Cayley has found that the squared coordinates x^2, y^2, z^2 of a point on the nodal curve can be expressed as rational functions of a single variable parameter σ.]

[445] A Memoir on Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69. Read February 10, 1870.]

The present Memoir is intended as a commencement of the theory of the quartic surfaces which have nodes (conical points). A quartic surface may be without nodes, or it may have any number of nodes up to 16. I show that this is so, and I consider how many of the nodes may be given points. Although it would at first sight appear that the number is 8, it is in fact 7; viz., we can, with 7 given points as nodes (but not in a proper sense with 8 or more given points), find a quartic surface; such surface contains in its equation 6 constants, which may be such that the surface has an additional node or nodes. Suppose that the surface has an 8th node: there are two distinct cases; viz., (1) the 8 nodes are the points of intersection of 3 quadric surfaces, or say they are an octad, and the surface is said to be octadic; (2) the 8th node is any point whatever on a certain sextic surface determined by means of the 7 given nodes, and called the dianodal surface of these 7 points; the quartic surface is said to be a dianome. The two cases are in general exclusive of each other; viz., the 7 given points being any points whatever, the dianodal surface does not pass through the 8th point of the octad; and thus the quartic surface with the 8 nodes is either octadic or else a dianome. Assuming it to be a dianome, the constants may be further determined so that there shall be a 9th node; it is necessary to examine whether this forms with 7 of the 8 nodes an octad. Supposing that it does not (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 9th node is then any point whatever on a certain curve of the order 18, determined by means of the 8 nodes, and called the dianodal curve of these 8 points. And, finally, the constants may be further determined so that there shall be a 10th node; supposing, as before, that this does not form an octad with any 7 of the 9 nodes (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 10th node is then any one of a system of 22 [should be 13] points determined by means of the 9 nodes, and called the dianodal system of these 9 points. But the quartic surface is now completely determined; viz., starting with any 7 given points as nodes, we have a dianome with 8 nodes, 9 nodes, or 10 nodes, say, an octodianome, enneadianome, or decadianome, but not with any greater number of nodes; these can only present themselves when particular conditions are satisfied in regard to the 7 given nodes, and to the 8th and 9th node; and the consideration of the quartic surfaces with more than 10 nodes would thus form a separate branch of the subject.

The case of the decadianome (or quartic surface with 10 nodes formed as above with 7 given points as nodes) is peculiarly interesting. I identify this with the surface which I call a symmetroid; viz., the surface represented by an equation $\Delta = 0$, where Δ is a symmetrical determinant of the 4th order the several terms whereof are linear functions of the coordinates (x, y, z, w); this surface is related to the Jacobian surface of 4 quadric surfaces (itself a very remarkable surface), and this theory of the symmetroid and the Jacobian, and of questions connected therewith, forms a large portion of the present Memoir.

The theory of the Jacobian is connected also with the researches in regard to nodal quartic surfaces in general; and, for greater clearness, it has seemed to me proper to commence the Memoir with certain definitions, &c., in regard to this theory. It will be seen in what manner I extend the notion of the Jacobian.

I remark that the present researches on Quartic Surfaces were suggested to me by Professor Kummer's most interesting Memoir "Ueber die algebraischen Strahlensysteme u.s.w.," *Berl. Abh.* 1866, in which, without entering upon the general theory, he is led to consider the quartic surfaces, or certain quartic surfaces, with $16, 15, 14, 13, 12$, or 11 nodes; the last of these, or surface with 11 nodes, being in fact a particular case of the symmetroid.

Considerations in regard to the Jacobian of four, or more or less than four, Surfaces.

1. In the case of any four surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, the differential coefficients of P, Q, R, S in regard to the coordinates (x, y, z, w) may be arranged as a square matrix in either of the ways

$$\begin{array}{c|cccc} & P, & Q, & R, & S \\ \hline \partial_x & & & & \\ \partial_y & & & & \\ \partial_z & & & & \\ \partial_w & & & & \end{array} \qquad \text{or} \qquad \begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \\ R & & & & \\ S & & & & \end{array}$$

according as we make P, Q, R, S the columns, or as we make them the lines, of the matrix; and we then have one and the same determinant, the Jacobian $J(P, Q, R, S) = 0$, regarding as zero the term in question.

2. In the case of more than four surfaces, adopting the arrangement

$$\begin{array}{c|cccccc} & P, & Q, & R, & S, & T, & \ldots \\ \hline \partial_x & & & & & & \\ \partial_y & & & & & & \\ \partial_z & & & & & & \\ \partial_w & & & & & & \end{array}$$

and considering the several determinants which can be formed with any four columns of the matrix, these equated to zero establish a more than one-fold relation between the coordinates, viz., in the case of five surfaces, we have a twofold relation, the intersection of the 5 surfaces in lines, instead of in points; and so in other cases. Or, what is the same thing, the Jacobians $J(P, Q, R, S)$, etc., equated to zero, are not independent equations.

3. In the case of fewer than four surfaces, adopting the arrangement

$$\begin{array}{c|ccc} & P, & Q, & R \\ \hline \partial_x & & & \\ \partial_y & & & \\ \partial_z & & & \\ \partial_w & & & \end{array}$$

and considering the several determinants which can be formed with any 3 of 4 columns of the matrix, we establish a more than ordinary relation, the simultaneous vanishing of J_1, J_2, J_3, J_4 formed by omitting in succession each of the rows; this gives a curve of intersection of higher order than the proper intersection. So in other cases. It will be seen in what manner I extend the notion of the Jacobian.

4. We say that a surface $P = 0$ contains an arbitrary point, or that the equation $P = 0$ is satisfied by the coordinates of an arbitrary point, when there is one or more relations between the coefficients of P, viz., P is regarded as a function of (x, y, z, w) and of a series of coefficients which enter linearly therein. To say that the surface passes through 1 given point is to assert one such relation; to say that it passes through 2 given points is to assert two such relations; and so on.

5. Suppose that the orders of the surfaces $P = 0$, $Q = 0$, ... are $a + 1$, $b + 1$, ... (so that the orders of the differential coefficients are a, b, ...); then the Jacobian $J(P, Q, R, S)$ is of the order $a + b + c + d$, and the corresponding surface has $(a + b + c + d)$-tic contact with the surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$ at the points of common intersection of the four surfaces.

6. It is also shown (Salmon, Ed. 2, p. 495) that the surface obtained by equating the discriminant of a quadric surface to zero is the Jacobian of the four polar quadric surfaces with respect to four arbitrary points, viz., the Jacobian determinant $J(P, Q, R, S)$, regarded as a function of (x, y, z, w), expresses the condition that the four planes of polarity should pass through one and the same point. So in other analogous statements about polars and Jacobians.

7. The subsidiary theorem of the contact of the curve and surface above mentioned is, that if for shortness the differential coefficients of P, Q, R, S in regard to (x, y, z, w) be denoted by P_x, ..., Q_x, ..., etc., the Jacobian $J(P, Q, R, S) = 0$ may be written, for x, y, z, w replaced by the coordinates of a point on the curve of intersection,

$$\begin{vmatrix} P_x, & Q_x, & R_x, & S_x \\ P_y, & Q_y, & R_y, & S_y \\ P_z, & Q_z, & R_z, & S_z \\ P_w, & Q_w, & R_w, & S_w \end{vmatrix} = 0,$$

and this determinant vanishes identically by virtue of the relation between the partial derivatives at the points common to the surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$. The contact is therefore of the order $a + b + c + d$ at the points of common intersection.

8. The subsidiary theorem of the contact of the curve and surface just mentioned gives rise to the Jacobian as the locus of points the polar planes of which, with respect to P, Q, R, S, pass through one and the same point. We shall in the sequel make considerable use of this theorem in various forms.

9. The Jacobian is then easily reduced to its proper form by a process of elimination, viz., introducing as many auxiliary equations as may be required to render it determinate.

10. We have, in particular, the theorem that the Jacobian of four quadric surfaces is a quartic surface, since each of the differential coefficients is of order 1 and there are four of them.

11. The Jacobian of four quadric surfaces is in fact a quartic surface, and the corresponding surface has 4-tic contact with the four quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$ at the points of common intersection.

12. In the case where there is a plane 1, the sextic cone breaks up into this plane and into a quintic cone, etc.

13. In the cases $k = 1, 2, 3, 4, 5$, and $k = 15, 16$, there is only one figure, and the number of arrangements may, on referring to the table, be at once seen.

Circumscribed Sextic Cone.

Order of Surface	
2	6
3	6, 1
4	6, 2, 1
5	6, 3, 1, 1
6	6, 4, 2, 1, 1
7	6, 5, 2, 1, 1, 1
8	6, 5, 3, 2, 1, 1, 1
9	6, 5, 4, 2, 1, 1, 1, 1
10	6, 5, 4, 3, 2, 1, 1, 1, 1
11	6, 5, 4, 3, 2, 1, 1, 1, 1, 1
12	... 6, 4, 2, 1, 1, 1, 1, 1, 1, 1, 1
13	... 5, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
14	... 3, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
15	... 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
16	... 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

and moreover, in the cases where there are two or more lines of the sextic cone, this the k sextic cones may be of the different forms in various combinations. The total number of cases, proper *facies* (*i.e.* this very gens), but not only a comparatively small number of them actually exist.

14. In the case where there is a plane 1, the sextic cone breaks up into this plane and into a (quintic or improper quintic cone intersecting this plane in 5 lines, etc.). There is, in fact, in the plane 1 a cubic, the plane is, in fact, a singular tangent plane meeting the surface in a conic twice repeated; and the 5 residual lines on the quintic cone touching the surface in a single point of the same kind (the line through the plane and the surface meeting this plane in three points each), and the like for the conic.

15. But a different form of expression is sometimes convenient: the conditions to be satisfied are frequently such that, being satisfied by the surfaces $P = 0$, $Q = 0$, they will be satisfied by the surface $\lambda P + \mu Q = 0$ where, for a given value of $\lambda : \mu$, there is a 1-tuple infinity of surfaces; or, in the same way, satisfied by the surfaces $P = 0$, $Q = 0$, $R = 0$ they will be satisfied by the surface $\lambda P + \mu Q + \nu R = 0$, where there is a 2-tuple infinity of surfaces; and so on.

16. Thus, for the quadric not subjected to any conditions, there are 9 constants (this is, 1 to fix the surface itself, and 8 in the equation thereof; the surface contains 9 constants); and we have therefore one and the same thing in the various forms in which the quartic surface in 9 nodes presents itself.

17. In the questions in which such quadric surfaces present themselves, we have to fix our ideas in regard to the number of constants. If, in such a question, there occur quadric surfaces $P = 0$, $Q = 0$, etc., we are to consider that we may through this or that function of P, Q, etc., have, for any given relations between the surfaces, so many constants; and the calculations are to be conducted accordingly.

18. A quartic surface contains 34 constants. We have, in fact, in the equation of a quartic surface

$$(a, b, c, d, e, f, g, h, i, j, k, l, m, n, p, q, r, s, t, u, v, w, x, y, z, \ldots)$$

35 coefficients, and so on. In a way which it is needless to recapitulate here, we ultimately arrive at the result that the quartic surface contains 34 constants.

19. In particular if we are concerned with a quartic surface having a node, then we may take the node as the point $(0, 0, 0, 1)$, viz., as the point of intersection of three of the four reference planes. The quartic surface then has the term in w^4 absent, etc.

20. The case of 4 given nodes is just as easy. In particular if we are concerned with a quartic surface having 4 nodes, then we may take these as the four points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, viz., as the four vertices of the tetrahedron of reference.

21. In proof, observe that, taking the 4 given nodes to be the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$ as before, the equation of a quartic surface having these four nodes contains a definite number of constants.

22. In the case of 5 given nodes, the number of constants should be reduced still further. We have through the 5 given points $\binom{5}{2} = 10$ lines, the common nodes; the remaining number of constants is $34 - 5 \cdot \text{constraints} = $ a definite count.

23. In verification, take the first 4 nodes to be as above, and let the 5th node be the point $(\alpha, \beta, \gamma, \delta)$.

24. In the case of 6 given nodes, the quartic surface should contain 10 constants. We have through the 6 given points $\binom{6}{2} = 15$ lines, and the calculation proceeds in the manner just indicated.

25. The 6 given nodes being any points whatever, their coordinates may be taken to be $(\alpha_1, \beta_1, \gamma_1, \delta_1), \ldots, (\alpha_6, \beta_6, \gamma_6, \delta_6)$; we may by a linear transformation take the first four of them to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$; and then the remaining two as $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$. The conditions that the quartic surface should have nodes at these 6 points reduce the number of constants to 10.

As to the Surface obtained by equating to zero a Symmetrical Determinant.

(continued). In the above (Salmon, Ed. 2, p. 493) that the surface obtained by equating to zero any symmetrical determinant has a determinate number of nodes; viz., if the order of the determinant is n, the number is $\binom{n+1}{3}$. In particular, the symmetrical determinant of the 4th order (Jacobian) has $\binom{5}{3} = 10$ nodes, as in the text of this Memoir.

26. The Jacobian is then easily found to be the symmetrical determinant

$$\Big| \partial_i \partial_j K \Big|_{i,j=1,\ldots,4} = 0,$$

that is, the corresponding surface is a quartic surface having a determinate number of nodes, viz., 10.

27. It may be shown *a posteriori* that J is not a quadric function of P, Q, R, S; and consequently the equation $J = 0$ represents a quartic surface, the Jacobian of the 4 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$.

28. The equation $J = 0$ is the locus of the vertices of the quadric cones which can be drawn through the curve of intersection of the surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$; that is, the locus of the vertices of the quadric cones which can be drawn through the 8 points of intersection; in particular, this Jacobian surface passes through the 6 lines 12, 34, 56 (viz., line of intersection of the planes through 1, 2, 3, and through 4, 5, 6); through 13, 24, 56; etc.

29. The surface in question (the Jacobian of the four quadric surfaces, or of the six points) is then a quartic surface having the six points as nodes; the order of the surface is 4 and the number of nodes is 10; we have, in fact, in the equation of the Jacobian surface $34 - 10 \cdot \text{some count} =$ the right number of constants.

30. In connexion with what precedes, we may here consider a curve which presents itself in connexion with the Jacobian surface above mentioned. We have, in fact, through any 7 given points $\binom{7}{2} = 21$ lines, through any 3 of which $4 \cdot \binom{7}{3}/3$ planes are determined; the Jacobian surface contains these lines as part of itself, together with a curve of certain order containing the points.

31. Suppose, however, that one of the seven points, instead of being a node of the dianodal surface, is itself a point on the dianodal surface; the question then arises whether the dianodal surface still passes through this point and whether the remaining structure is preserved.

32. In the case where two of the points are each of them the vertex of a cone passing through the other 5 points, the dianodal surface degenerates further: one or more of the conic relations comes to a duplication.

33. *Particular cases of the dianodal surface.* There are many particular cases of the dianodal surface, but it is unnecessary to enumerate them; in fact, throughout the general theory, the form of the dianodal surface remains the same.

34. It follows that, if $\Delta = 0$ be another quartic surface having the 7 given nodes, then the equation of the dianodal surface (in regard to the symmetrical determinant) must contain the function of Δ, etc.

35. A particular quartic surface having (in an improper sense) the 7 given nodes, viz., in many ways, but in special cases as follows: take any plane $M = 0$ through three of the nodes, say through 1, 2, 3; and any cubic surface $\Omega = 0$ having the other 4 nodes, viz., 4, 5, 6, 7; then the product $M \cdot \Omega$ vanishes at each of the 7 given points.

36. We can with the 7 given points form 35 such combinations $M \cdot \Omega$; these are of course equivalent in virtue of the dianodal relations among them.

37. It has already been shown that a quartic surface cannot in a proper sense have 8 given nodes. In regard to the quartic surfaces with 8 nodes, we start from the dianodal surface as described, and add an 8th node.

38. But if θ be not $= 0$, there may still be an 8th node; viz., this must then lie on the dianodal surface, etc.

39. I say that the 8th node may be any point whatever on the dianodal surface, that is, the dianodal surface is the locus of the 8th node.

40. [*Article 40 develops further the parametric form of the dianodal surface and its relation to the* 7 *given nodes.*]

41. [*Article 41 contains supplementary remarks bearing on the same construction.*]

42. Consider the seven points $1, 2, 3, 4, 5, 6, 7$. As already mentioned, through any 3 of them, say $1, 2, 3$, we have a plane $M = 0$; and through the same 7 points, with the 4 points $4, 5, 6, 7$ as nodes, a cubic surface $\Omega = 0$. We may take the planes through 123, 145, and 456 respectively each as M, and the cubic surfaces having Ω as nodes the remaining 4 points; and similarly, taking other combinations, we have $\Delta = M\Omega = 0$, a quartic surface with the seven points as nodes. And using this form of Δ it may be shown that the dianodal $J(P, Q, R, \Delta) = 0$ passes through the 21 lines $12, 13, \ldots 67$, and through 35 plane cubics such as $M = 0$, $\Omega = 0$; viz., this is a cubic in the plane 123 passing through the points $1, 2, 3$, and through the intersection of the plane with each of the six lines $45, 46, \ldots 67$ (nine points determining the cubic); the complete intersection by the plane 123 being therefore composed of this cubic and of the three lines $12, 13, 23$. For the passage through the cubic, we have only to observe that

$$J(P, Q, R, M\Omega) = J(P, Q, R, \Omega)\, M + J(P, Q, R, M)\, \Omega = 0$$

is satisfied by $M = 0$, $\Omega = 0$; and for the passage through the lines, taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the planes $567, 674, 745$, and 456 respectively, each of the functions P, Q, R is of the form $ayz + bzx + cxy + fxw + gyw + hzw$, and the function Ω is of the form $Ayzw + Bzwx + Cwxy + Dxyz$. Hence, writing in the derived functions $z = 0$, $w = 0$, the first and second lines of the determinant $J(P, Q, R, \Omega)$ will be of the form

$$\begin{pmatrix} c\,y, & c'\,y, & c''\,y, & * \\ c\,x, & c'\,x, & c''\,x, & * \end{pmatrix},$$

or the determinant vanishes for $z = 0$, $w = 0$; that is for any point of the line 45 we have $\Omega = 0$ and also $J(P, Q, R, \Omega) = 0$; consequently $J(P, Q, R, M\Omega) = 0$, and the like for the other lines. The theorem is thus proved.

43. I say that the dianodal surface passes through each of the 7 skew cubics, such as 123456. To prove this, it is only necessary to show that the skew cubic 123456 lies on the dianodal surface. For this purpose it will be enough to show that the skew cubic meets the plane 712 in a point of the surface; for then it will, in like manner, meet each of the 15 planes $712, 713, \ldots 756$ in a point of the surface; that is, we shall have 15 intersections of the curve and surface, and there are, besides, the intersections $1, 2, 3, 4, 5, 6$, in all 21 intersections; that is, the skew cubic must lie on the surface.

44. The plane 712 meets the surface in three lines and in a plane cubic determined by the points $7, 1, 2$ and the six intersections of the plane with the lines $34, 35, \dots 56$. We have therefore to show that this plane cubic meets the skew cubic 123456. Consider for a moment the points $1, 2, 3, 4, 5, 6$ and another point $7'$. As seen above, we have in general, through the points $1, 2, 7'$ and with the points $3, 4, 5, 6$ as nodes, a determinate cubic surface, which surface passes through the lines $34, 35, \dots 56$. But the cubic surface becomes indeterminate if the points $1, 2, 7', 3, 4, 5, 6$ are on the same skew cubic; that is, if $7'$ is any point whatever on the skew cubic 123456 (the proof presently). Taking, then, $7'$ as the intersection of the skew cubic by the plane 712, we have in this plane the points $7', 1, 2$, and the intersections of the plane by the lines $34, 35, \dots 56$, nine points through which there pass an infinity of plane cubics; that is, the plane cubic determined by the points $7, 1, 2$ and the six intersections will pass through the point $7'$; viz., it meets the skew cubic 123456.

45. For the subsidiary theorem, taking X, Y, Z, W as current coordinates, viz., $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ as the equations of the planes $456, 563, 634, 345$ respectively, (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) as the coordinates of the points 1 and 2 respectively, and (x, y, z, w) for those of $7'$; the equation of the cubic surface passing through $7', 1, 2$, and having the nodes $3, 4, 5, 6$, is

$$\begin{vmatrix} \frac{1}{X}, & \frac{1}{Y}, & \frac{1}{Z}, & \frac{1}{W} \\ \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

and this ceases to be a determinate function if only

$$\begin{vmatrix} \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

viz., considering (x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2) as given, this is a twofold relation between the coordinates (x, y, z, w) of the point $7'$. The relation may be represented by the four equations $(yzw) = 0$, $(zwx) = 0$, $(wxy) = 0$, $(xyz) = 0$, if for shortness

$$(yzw) = \begin{vmatrix} yz, & zw, & wy \\ y_1z_1, & z_1w_1, & w_1y_1 \\ y_2z_2, & z_2w_2, & w_2y_2 \end{vmatrix}$$

and the like as to the other symbols. The four equations represent quadric surfaces, each two intersecting in a line [e.g., $(yzw) = 0$, $(zwx) = 0$ in the line $z = 0$, $w = 0$], and the four surfaces besides intersecting in a skew cubic, which is the required locus of the point $7'$, and which, as is seen at once, passes through the points $1, 2, 3, 4, 5, 6$.

46. By what precedes, we have on the dianodal surface through the point 1 the lines $12, 13, 14, 15, 16, 17$, and the skew cubics 123456, &c. The six lines are not on the same quadric cone, and it thus appears that the point 1 must be a cubic-node (point where, instead of the tangent plane, we have a cubic cone) on the surface. It is to be remarked that the lines $12, 13, 14, 15, 16$, and the tangent at 1 to the skew cubic 123456, lie in a quadric cone; viz., this tangent is given as the sixth intersection of the cubic cone with the quadric cone through the lines $12, 13, 14, 15, 16$.

47. I revert to the equation of the dianodal surface as given in the form $J = J(P, Q, R, M\Omega) = 0$, where $M = 0$ is the plane through the points $1, 2, 3$, and $\Omega = 0$ the cubic surface through these points, and having the points $4, 5, 6, 7$, as nodes. We can find the orders of the several functions P, Q, R, M, Ω in the coordinates (x_1, y_1, z_1, w_1), &c., of the several points; viz., writing for shortness x_1^2 to denote the order 2 in regard to (x_1, y_1, z_1, w_1), and so in other cases, we have

$$P = Q = R = x^2(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^2,$$

$$M = x(x_1, x_2, x_3),$$

$$\Omega = x^3(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^3,$$

(where, of course, the x's show in like manner the orders in regard to the current coordinates (x, y, z, w); the proof in regard to Ω is easily supplied.) The order of J is equal that of $PQRM\Omega$, less 4 as regards the current coordinates, by reason of the differentiations; that is, we have $J = x^6(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$; and we thus see that the equation of the dianodal surface as above obtained is encumbered with a constant factor of the form $(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$. In fact, the relation between the 7 points and the current point (x, y, z, w), or say the point 8, as expressing that the 8 points are the nodes of a dianome, should be a symmetrical one in regard to the coordinates of the several points; and being of the order 6 in regard to the coordinates (x, y, z, w), it should be of the same order in regard to the other coordinates; that is, the true form would be $J = (x\, x_1\, x_2\, x_3\, x_4\, x_5\, x_6\, x_7)^6 = 0$.

48. It is possible that taking the 4 points, say $1, 2, 3, 4$, to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, and the 3 points, say $5, 6, 7$, to be $(1, 1, 1, 1)$, $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, the extraneous factor might exhibit itself, and that the calculation thereby be simplified. But I do not at present further pursue the inquiry.

Dianodal Curve of 8 given Points, or say a Nodo or Section.

49. The dianodal surface, qua surface having the last-mentioned 8 points for nodes, has on it various special curves, and the discussion of these is here in some sort necessary; viz., we have on it a curve, the section by an arbitrary plane, of the order 6; the curve of intersection of the dianodal surface with another quartic surface through the 7 given nodes and called the dianodal curve of these 8 points. There exists between $P, Q, R, S = M\Omega$ a syzygetic relation

$$\alpha P + \beta Q + \gamma R + \delta\, M\Omega = 0,$$

where in $\alpha, \beta, \gamma, \delta$ are linear functions of the current coordinates, α for instance being a function which vanishes at the points $1, 2, 3$, and which contains a number of arbitrary coefficients; whence the syzygetic relation is one satisfied at every one of the 7 given nodes.

50. This being so, the cubic curve through the last-mentioned six points has its equation of the form $J(P, Q, R, \alpha M\Omega) = 0$; and the dianodal curve of these 8 points is the intersection of the dianodal surface with this cubic curve.

51. Considering the orders in regard to $(a, b, c, d, f, g, h, l, m, n, P, Q, R, S, T, U, V)$, the dianodal curve through the points $1, 2, \ldots, 7$, and an 8th point lies on the dianodal surface, and is of the order 18. It contains in fact one set of intersections (the 21 lines $12, 13, \ldots, 67$ counted as one with multiplicity 2) and the curve proper; this last is the dianodal curve of the 8 points.

52. We have $5 \cdot 6 = 30$ as the order of intersection of the dianodal surface (sextic) and the auxiliary quartic surface; the auxiliary quartic passing through the 7 given points counts as $\binom{7}{2} = 21$, and the dianodal curve is of the order 18. There would be a reduction of 25 if the line through the 6 given points $1, 2, 3, 4, 5, 6$ were in involution with the 6 given quadric surfaces; whence the dianodal curve order would be $30 - 25 = 5$. But this is not the case.

53. In order to find it, taking as above 9 a given point on the Jacobian, we have a tenth point on the surface only if there be a relation between the coordinates of the points 1 through 9. The condition is in fact a determinate condition which fixes the order of the dianodal curve at 18.

54. If there is a 10th node, say 10, this is also a point on the Jacobian curve of the 8 points $1, 2, \ldots, 8$, and similarly a point on the dianodal curves through suitable octads of the 9 remaining nodes.

55. In the case of the surface with 9 nodes, it is clear that this is octadic in one way only: the node 9 cannot form an octad with any 7 of the remaining nodes. But in the case of the surface with 10 nodes, the question arises whether the nodes 9 and 10 may not be such as to form an octad with some six, say with the nodes $1, 2, 3, 4, 5, 6$ of the remaining 8 nodes; that is, whether we can have $1, 2, 3, 4, 5, 6, 7, 8$ forming an octad, and also $1, 2, 3, 4, 5, 6, 9, 10$ forming an octad. I will show that this is impossible if only the points $1, 2, 3, 4, 5, 6$ are given points, that is, points assumed at pleasure and not specially related to each other. For this purpose, assuming that the points form 2 octads as above, take through $1, 2, 3, 4, 5, 6, 7, 9$ the quadric surfaces $P = 0$, $Q = 0$, then each of these passes through 8, 10; take $R = 0$ any other quadric surface through $1, 2, 3, 4, 5, 6, 7, 8$, and $S = 0$ any other quadric surface through $1, 2, 3, 4, 5, 6, 9, 10$. Then $P = 0$, $Q = 0$, $R = 0$ intersect in the 1st octad, [*continues in book p. 154; the memoir then resumes at* § 60 *in the separate retype on book p. 155.*]

(End of chunk: book pages 129–153 of Vol. VII. The memoir on Quartic Surfaces continues with §§ 56–59 on book p. 154, and is then resumed at § 60 in `cayley_vol07_paper_445_RETYPE.tex`*.)*